# FERA ফেরা

D5 / CONTRACTS · VERSION 2.0

# Background Schema & Registry

Shared TypeScript contracts, domain relationships and 537 IDs

**Specification handover · Implementation unverified**

**Authors**

Sadman Shaharier · Obidit Islam

Issued 11 October 2026 · Asia/Dhaka (UTC+6)

Invented synthetic-data prototype · Interactive team edition

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> Click a section or use the PDF bookmarks. Requirement IDs link to their definitions. External sources are clickable. The meeting workbook contains blank fillable review fields. Application acceptance and pilot evidence remain pending.


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D5 / CONTRACTS / V2.0

# Background Schema & Registry

Shared TypeScript contracts, domain relationships and 537 IDs · Authors: Sadman Shaharier and Obidit Islam

↗ Contents

| Field | Value |
|---|---|
| Version | 2.0 |
| Date | 11 Oct 2026 (Asia/Dhaka, UTC+6) |
| Owner | Sadman Shaharier |
| Authors | Sadman Shaharier; Obidit Islam |
| Status | V2 team handover; implementation status UNVERIFIED |
| Companion | `fera_types_v2.ts` — byte-identical copy of §6 |
| Covers | All nine V2 source documents and the three V2 companions; validation report. |

## V2 release contract

**Authors:** Sadman Shaharier and Obidit Islam. **Version:** 2.0. **Issued:** 11 October 2026, Asia/Dhaka (UTC+6).

This is the revised team specification, not evidence that the application has been built. Actual repository progress, account configuration, interviews, evaluations, and spend remain **UNVERIFIED** until evidence is attached. V1 defaults are superseded by the V2 rules below; existing requirement IDs are retained.

- This release is a **synthetic-data prototype**. Officer verification and referral delivery are simulated and labelled. The schema rejects records marked non-synthetic; the application must also prevent real reports from entering this release.
- A candidate score is similarity, not identity probability. A reviewer decision does not establish guardian authority or safe reunification. A referral row means **referral recorded**, not delivered or acknowledged.
- Draft payload stays on the device before report confirmation, but providers process audio/text beforehand with a notice. Only expiry/counter/hash metadata and cost reservations are stored server-side before submission.
- Possible duplicates remain visible. Shared phone alone never establishes one child. No physical child-return action is performed by Fera.
- Counts distinguish missing-report rows, found-report rows, and confirmed pairs. They are not a census of unique children or proof of rescue.
- Sources, diagrams, tables, code blocks, and acceptance criteria are part of the handover. The companion code must match the embedded code exactly. Team handover records change closure, evidence gaps, and release gates.

# Changes to earlier documents

V2 consolidates 537 unique IDs, their corrected definition locations, new API/entity/test/task records, and the revised TypeScript contracts. All companion snippets match their delivered files. D2A and D5 consistently use Uint8Array at the speech adapter boundary.

# 1 Project metadata

| Field | Value |
|---|---|
| Name | Fera (ফেরা) |
| One line | Bangla-first reviewed report and candidate coordination; no rescue guarantee |
| Event | Hack for Humanity 2026 (AI Collective Bangladesh, AIUB), Track 3 Smart Society & Public Services |
| Round 2 due | 14 Oct 2026 confirmed date; exact time/deliverables/submission channel pending organiser email |
| Grand Finale | 24 Oct 2026 at AIUB: 5–10 min presentation + live demo |
| Judging | Social/human impact, feasibility, scalability; weights UNKNOWN (OQ-025); no equal-weight claim |
| Team | Sadman Shaharier (lead/integration), Obidit Islam (intake & LLM), Masnun Kabir (data, matching & dashboard) — institutions/registration spelling pending verification |
| Platform | Responsive web app; mobile-first for parents, desktop-first for officers; no hardware |

# 2 Problem context

| Item | Value and evidence status |
|---|---|
| Reported missing, Jan–Jul 2026 | 15,717 children |
| Still untraced | 2,038 (13 %), Jan–Jul cohort reported 3 Sep 2026; 13,679 were reported returned/rescued. No claim that existing agencies record no outcomes. |
| Aged 10–17 | 97 % |
| Rate | About 74 per day |
| Linked operational registry | Unverified hypothesis; no absence claim |
| Existing system | CID's MUN Alert, live since 13 Jan 2026, helpline 13219, joined Meta's alert network May 2026 |

Breaking points: lost first hours; scattered reports with no missing↔found link; unvalidated need for clearer case-level coordination follow-up.

# 3 Domain concepts

| Concept | Definition in Fera | Fera's role |
|---|---|---|
| GD (General Diary) | Police-station record of a reported incident, here a missing child | Fera drafts the text; the family files it (FR-008) |
| MUN Alert | CID's missing-child alert system with Amber Alert for Bangladesh | Fera never issues or withdraws alerts (BR-002) |
| 13219 | MUN Alert's toll-free helpline | Fera shows the number and a summary to read out (FR-010) |
| CID Missing Children Cell | Unit named in sources as able to issue or withdraw alerts; public briefing names Missing Children Cell; operational partnership unverified | External; no integration |
| Missing case | Report of a child not found by family | `cases.kind = missing` |
| Found case | Report of a child found by staff, shelter, station or public source | `cases.kind = found` |
| Match | A ranked missing↔found pair; a candidate until an officer decides | ENTITY-005; BR-001 |
| Referral | Internal simulated referral record after a reviewer decision; not proof of delivery, custody, identity or safe resolution | ENTITY-007; BR-005 |
| Duplicate | Two same-kind records likely describing one child | ENTITY-008; BR-021 |
| Closure | A missing case reaching `closed` with an outcome | FR-018; GOAL-006 |
| Disaster mode | Planned extension for flood/cyclone shelters; no measured need | FR-036 (Could) |
| Synthetic data | Team-written records with known ground truth, labelled on screen | **BR-026** |

# 4 Stakeholders

| Stakeholder | Interest | Interaction |
|---|---|---|
| Parents and guardians (PERSONA-001) | Fast, complete report; updates | Parent screens |
| Verifying officers (PERSONA-002) | Accurate shortlist; accountable decisions | Officer screens |
| Shelter/station staff (PERSONA-003) | Found children matched | Staff screens |
| Child-protection caseworkers (PERSONA-004) | Safe placement after a match | Referral record only |
| Disaster-shelter volunteers (PERSONA-005) | Reunification after floods/cyclones | Could, SCREEN-015 |
| CID / MUN Alert | Cleaner cases; alert authority | No integration; future data agreement |

| Stakeholder | Interest | Interaction |
|---|---|---|
| Organisers, judges, mentors | Impact, feasibility, scalability | Deck, video, demo |
| AI providers | Process audio and report text | Via adapters (TB4) |
| Team | Deliver by 14 and 24 Oct | All |

# 5 Constraints

| Constraint | Value |
|---|---|
| AI budget | USD 30 total, hard cap USD 25 in app (DECISION-017) |
| Time | Round 2 on 14 Oct; exact official cutoff pending; finale 24 Oct |
| Capacity | Actual remaining capacity pending; at most 120 h for a fresh four-day start at assumed 10 h/person/day (A-040) |
| Network | Venue internet may fail; demo-safe mode required |
| Devices | Laptop + Android phone browser |
| Language | Bangla text and voice end to end; mixed script and Bangla numerals |
| Data | No private data; no scraping of private accounts; synthetic only in this release; public real-child material requires a later lawful-use gate |
| Safeguarding | BR-001 – BR-011 non-negotiable |
| Law | Personal Data Protection Act, 2026, qualified interpretation and effective provisions review required before real-child use; no compliance claim (OQ-011) |

# 6 TypeScript interfaces

`fera_types_v2.ts` is a byte-identical copy of this block. It type-checks with `tsc --strict` (TypeScript 5.4).

```
// Fera shared types V2.0 - D5 section 6. Exact companion of the V2 embedded block.
// Authors: Sadman Shaharier; Obidit Islam. 11 October 2026, Asia/Dhaka.
// Registry records (suite IDs) + domain records (D2 schema) + AI/API contracts (D2 §2, §5).

/* ---------- Shared primitives ---------- */
export type Uuid = string;
export type IsoDateTime = string; // ISO 8601 with +06:00
export type MoSCoW = "Must" | "Should" | "Could" | "Won't";
export type DocId = "00" | "D1" | "D2A" | "D2B" | "D3" | "D4" | "D6" | "D7";
export type Level = "L" | "M" | "H" | "Severe";
export type Role = "parent" | "staff" | "officer" | "admin";
export type AuthLabel = Role | "public";
export type Owner = "S" | "O" | "M";
```

```
/* ---------- Registry records ---------- */
export interface RegistryRecord {
  id: string;
  label: string;
  definedIn: DocId;
}
export interface Goal extends RegistryRecord {
  id: `GOAL-${string}`;
  metric: string;
  target: string;
  method: string;
  timeframe: string;
}
export interface Persona extends RegistryRecord {
  id: `PERSONA-${string}`;
  priority: MoSCoW;
}
export interface Story extends RegistryRecord {
  id: `STORY-${string}`;
  priority: MoSCoW;
  acceptance: string[];
  dependencyIds: string[];
  screenIds: string[];
}
export interface FunctionalRequirement extends RegistryRecord {
  id: `FR-${string}`;
  priority: MoSCoW;
  storyIds: string[];
  testIds: string[];
  mock: boolean;
}
export interface NonFunctionalRequirement extends RegistryRecord {
  id: `NFR-${string}`;
  category: string;
  measure: string;
}
export interface BusinessRule extends RegistryRecord {
  id: `BR-${string}`;
  safeguarding: boolean;
  frIds: string[];
  testIds: string[];
}
export interface Assumption extends RegistryRecord {
  id: `A-${string}`;
}
export interface Dependency extends RegistryRecord {
  id: `DEP-${string}`;
  owner: string;
  neededBy: string;
}
export interface Risk extends RegistryRecord {
  id: `RISK-${string}`;
  likelihood: Level;
  impact: Level;
  mitigation: string;
  owner: string;
}
export interface OpenQuestion extends RegistryRecord {
```

```
  id: `OQ-${string}`;
  defaultApplied: string | null;
  answeredBy: string;
  dueBy: string;
}
export interface Screen extends RegistryRecord {
  id: `SCREEN-${string}`;
  priority: MoSCoW;
  roles: Role[];
}
export interface Component extends RegistryRecord {
  id: `COMP-${string}`;
  layer: "server" | "client" | "ui";
}
export interface ApiEndpoint extends RegistryRecord {
  id: `API-${string}`;
  method: "GET" | "POST" | "PATCH" | "*";
  path: string;
  auth: AuthLabel[];
  priority: MoSCoW;
}
export interface EntityRecord extends RegistryRecord {
  id: `ENTITY-${string}`;
  table: string;
}
export interface Decision extends RegistryRecord {
  id: `DECISION-${string}`;
}
export interface TestCase extends RegistryRecord {
  id: `TC-${string}`;
  covers: string[];
  steps: string[];
  passIf: string;
}
export interface Task extends RegistryRecord {
  id: `TASK-${string}`;
  priority: MoSCoW;
  owners: Owner[];
  dependsOn: string[];
  hours: number;
  criticalPath: boolean;
}
export interface ErrorCode extends RegistryRecord {
  id: `E-${string}`;
  key: string;
  http: number | null;
  messageBn: string;
  messageEn: string;
}
export interface FlowAction extends RegistryRecord {
  id: `ACT-${string}`;
  roles: Role[];
}
export interface FlowDecision extends RegistryRecord {
  id: `D-${string}`;
}
export interface TraceLink {
  from: string;
  to: string;
```

```
  kind: "goal-story" | "story-fr" | "fr-screen" | "fr-test" | "br-fr" | "br-test" | "fr-task";
}

/* ---------- Domain enums (D2 §7) ---------- */
export type MatchingState = "queued" | "running" | "complete" | "failed";
export type MissingStatus = "submitted" | "open" | "matched_pending_review" | "confirmed" |
"closed";
export type FoundStatus = "open" | "matched_pending_review" | "referred";
export type CaseKind = "missing" | "found";
export type CaseStatus = "submitted" | "open" | "matched_pending_review" | "confirmed" | "closed" |
"referred";
export type StatusGroup = "open" | "matched" | "closed";
export type Sex = "male" | "female" | "other" | "unknown";
export type CaseSource = "parent" | "staff" | "public_import" | "synthetic";
export type UserRole = "staff" | "officer" | "admin";
export type CandidateStatus = "pending" | "confirmed" | "rejected" | "voided" | "superseded";
export type DecisionType = "confirm" | "reject";
export type Outcome = "coordination_recorded" | "resolved_outside_fera";
export type DuplicateStatus = "flagged" | "confirmed_same" | "dismissed";
export type OutputType = "gd" | "poster" | "helpline";
export type ActorType = "parent" | "staff" | "officer" | "admin" | "system";

/* ---------- Domain records (one per table) ---------- */
export interface User { // ENTITY-001
  id: Uuid;
  username: string;
  display_name: string;
  password_hash: string;
  role: UserRole;
  verified: boolean;
  district_code: number | null;
  is_active: boolean;
  created_at: IsoDateTime;
  updated_at: IsoDateTime;
}
export interface ParentSession { // ENTITY-002
  id: Uuid;
  created_at: IsoDateTime;
  last_seen_at: IsoDateTime;
  expires_at: IsoDateTime;
  revoked_at: IsoDateTime | null;
}
export interface Case { // ENTITY-003
  id: Uuid;
  ref_code: string;
  kind: CaseKind;
  status: CaseStatus;
  source: CaseSource;
  source_url: string | null;
  is_synthetic: boolean;
  reporter_session_id: Uuid | null;
  created_by_user_id: Uuid | null;
  raw_text: string | null;
  child_name: string | null;
  child_name_latin: string | null;
  age_years: number | null;
  age_min: number | null;
  age_max: number | null;
  sex: Sex;
```

```
  height_cm: number | null;
  clothing: string | null;
  marks: string | null;
  circumstances: string | null;
  event_at: IsoDateTime | null;
  event_at_approx: boolean;
  event_at_lower: IsoDateTime | null;
  event_at_upper: IsoDateTime | null;
  provenance: Record<string, EvidenceSpan>;
  embedding_model: string | null;
  embedding_revision: string | null;
  embedding_space: string | null;
  matching_state: MatchingState;
  matched_at: IsoDateTime | null;
  match_error_id: string | null;
  place_text: string | null;
  district_code: number | null;
  thana: string | null;
  reporter_relation: string | null;
  contact_phone: string | null;
  gd_number: string | null;
  embedding: number[] | null;
  client_submission_id: Uuid | null;
  submission_payload_hash: string | null;
  input_hash: string | null;
  version: number;
  outcome: Outcome | null;
  outcome_note: string | null;
  closed_at: IsoDateTime | null;
  anonymised_at: IsoDateTime | null;
  created_at: IsoDateTime;
  updated_at: IsoDateTime;
}
export interface CaseVersion { // ENTITY-004
  id: Uuid;
  case_id: Uuid;
  version: number;
  snapshot: Record<string, unknown> | null;
  edited_by: ActorType;
  edited_at: IsoDateTime;
}
export interface MatchCandidate { // ENTITY-005
  id: Uuid;
  missing_case_id: Uuid;
  found_case_id: Uuid;
  missing_version: number;
  found_version: number;
  score: number;
  rank: number;
  reason_bn: string;
  reason_en: string;
  components: MatchComponents;
  status: CandidateStatus;
  created_at: IsoDateTime;
  decided_at: IsoDateTime | null;
}
export interface MatchComponents {
  cosine: number;
  name: number;
```

```
  clothing: number;
  district: number;
  age: number;
}
export interface ReviewDecision { // ENTITY-006
  id: Uuid;
  candidate_id: Uuid;
  officer_id: Uuid;
  decision: DecisionType;
  officer_verified_at_decision: true;
  reason: string | null;
  decided_at: IsoDateTime;
}
export interface Referral { // ENTITY-007
  id: Uuid;
  candidate_id: Uuid;
  found_case_id: Uuid;
  missing_case_id: Uuid;
  agency_name: string;
  referred_by: Uuid;
  referred_at: IsoDateTime;
  record_state: "recorded";
  is_simulated: true;
  note: string | null;
}
export interface DuplicateLink { // ENTITY-008
  id: Uuid;
  case_a: Uuid;
  case_b: Uuid;
  score: number;
  rule: "profile" | "same_phone";
  status: DuplicateStatus;
  created_at: IsoDateTime;
  resolved_by: Uuid | null;
  resolved_at: IsoDateTime | null;
  reason: string | null;
}
export interface AuditEntry { // ENTITY-009
  id: number;
  at: IsoDateTime;
  actor_type: ActorType;
  actor_user_id: Uuid | null;
  actor_session_id: Uuid | null;
  action: string;
  case_id: Uuid | null;
  candidate_id: Uuid | null;
  details: Record<string, unknown>;
  request_id: string | null;
}
export interface District { // ENTITY-010
  code: number;
  name_en: string;
  name_bn: string;
  division_en: string;
  lat: number;
  lng: number;
}
export interface DistrictNeighbour { // ENTITY-011
  district_code: number;
```

```
  neighbour_code: number;
}
export interface GeneratedOutput { // ENTITY-012
  id: Uuid;
  case_id: Uuid;
  type: OutputType;
  text_bn: string | null;
  case_version: number;
  created_at: IsoDateTime;
}
export interface AiCall { // ENTITY-013
  id: number;
  at: IsoDateTime;
  draft_id: Uuid | null;
  case_id: Uuid | null;
  reservation_id: Uuid | null;
  request_id: string | null;
  operation: "transcribe" | "extract" | "embed";
  provider: string;
  model: string;
  duration_ms: number;
  input_units: number | null;
  output_units: number | null;
  est_cost_usd: number;
  cache_hit: boolean;
  ok: boolean;
  error_id: string | null;
}
export interface RateLimitBucket { // ENTITY-014
  key: string;
  window_start: IsoDateTime;
  count: number;
}

// New V2 metadata entities; these store no child text or audio before submission.
export interface IntakeDraft { // ENTITY-015
  id: Uuid;
  session_id: Uuid | null;
  user_id: Uuid | null;
  kind: CaseKind;
  created_at: IsoDateTime;
  expires_at: IsoDateTime;
  payload_hash: string | null;
  ai_calls_used: number;
  revision: number;
}
export interface AiBudget { // ENTITY-016
  id: true;
  cap_usd: number;
  committed_usd: number;
  reserved_usd: number;
}
export interface AiReservation { // ENTITY-017
  id: Uuid;
  draft_id: Uuid | null;
  reserved_usd: number;
  actual_usd: number | null;
  state: "reserved" | "settled";
  created_at: IsoDateTime;
```

```
  settled_at: IsoDateTime | null;
}
export interface EvidenceSpan {
  quote: string;
  start_utf16: number;
  end_utf16: number;
  derivation: "direct" | "digits" | "time_interval" | "district_mapping" | "romanisation";
}

/* ---------- AI adapter contracts (D2 §2.1–§2.3) ---------- */
export interface ExtractInput {
  kind: CaseKind;
  sourceText: string; // reviewed raw transcript/text; evidence quotes refer to this
  normalisedText: string; // deterministic normalisation; preserve span mapping
  nowIso: IsoDateTime;
}
export interface ExtractOutput {
  child: {
    name: string | null;
    age_years: number | null;
    age_min: number | null;
    age_max: number | null;
    sex: Sex;
    height_cm: number | null;
    clothing: string | null;
    marks: string | null;
  };
  event: {
    at: IsoDateTime | null;
    at_approx: boolean;
    at_lower: IsoDateTime | null;
    at_upper: IsoDateTime | null;
    place_text: string | null;
    district_text: string | null;
    thana: string | null;
    circumstances: string | null;
  };
  reporter: { relation: string | null; phone: string | null };
  evidence: Record<string, EvidenceSpan>;
}
export interface LLMProvider {
  readonly id: "gemini" | "openai" | "claude";
  extractCase(input: ExtractInput): Promise<ExtractOutput>;
}
export interface SpeechProvider {
  readonly id: "whisper" | "cache";
  transcribe(audio: Uint8Array, mime: string, langHint: "bn"): Promise<{ text: string; durationS:
number }>;
}
export interface EmbeddingProvider {
  readonly id: "local" | "api";
  readonly dim: 384;
  embed(texts: string[]): Promise<number[][]>;
}

/* ---------- API shapes (D2 §5) ---------- */
export interface CaseDraft extends ExtractOutput {
  kind: CaseKind;
  source_text: string; // browser payload; persist only after explicit report confirmation
```

```
  child: ExtractOutput["child"] & { name_latin: string | null };
  event: ExtractOutput["event"] & { district_code: number | null };
}
// Narrow DTOs, not ORM rows. Server projections are explicit allowlists.
export interface OwnerCaseView {
  id: Uuid; ref_code: string; kind: "missing"; status: MissingStatus;
  status_group: StatusGroup; is_synthetic: true; version: number;
  editable: boolean; matching_state: MatchingState;
  draft: CaseDraft | null; // own editable report only; null after terminal lock/redaction
  created_at: IsoDateTime; updated_at: IsoDateTime;
}
export interface StaffOwnCaseView {
  id: Uuid; ref_code: string; kind: "found"; status: FoundStatus;
  is_synthetic: true; version: number; matching_state: MatchingState;
  draft: CaseDraft | null; created_at: IsoDateTime;
}
export interface AdminCaseView {
  id: Uuid; ref_code: string; kind: CaseKind; status: CaseStatus;
  district_code: number | null; is_synthetic: true;
  anonymised_at: IsoDateTime | null;
  // No child payload, contact, candidate IDs/scores, or free-text notes.
}
export interface OfficerCaseView {
  id: Uuid;
  ref_code: string;
  kind: CaseKind;
  status: CaseStatus;
  status_group: StatusGroup;
  is_synthetic: boolean;
  source: CaseSource;
  source_url: string | null;
  child: {
    name: string | null;
    age_years: number | null;
    age_min: number | null;
    age_max: number | null;
    sex: Sex;
    height_cm: number | null;
    clothing: string | null;
    marks: string | null;
  };
  event: {
    at: IsoDateTime | null;
    at_approx: boolean;
    at_lower: IsoDateTime | null;
    at_upper: IsoDateTime | null;
    place_text: string | null;
    district_code: number | null;
    district_name_bn: string | null;
    thana: string | null;
  };
  contact_phone: string | null; // verified officer projection only; never logged
  contact_phone_masked: string | null;
  version: number;
  duplicate_of: string[]; // review cues; flagged links never suppress a distinct report
  matching_state: MatchingState;
  created_at: IsoDateTime;
  updated_at: IsoDateTime;
}
```

```
export interface NextQuestion {
  field: string;
  bn: string;
  en: string;
}
export interface ExtractResponse {
  draft_token: string; // opaque signed draft/principal/expiry binding; usage is server-owned
  draft_id: Uuid;
  draft_revision: number;
  expires_at: IsoDateTime;
  draft: CaseDraft;
  missing_fields: string[];
  next_question: NextQuestion | null;
  ai_calls_used: number;
  cached: boolean;
}
export interface TranscribeResponse {
  draft_token: string;
  draft_id: Uuid;
  ai_calls_used: number;
  transcript: string;
  language: "bn" | "mixed";
  duration_s: number;
  cached: boolean;
}
export interface OutputResponse {
  type: OutputType;
  text_bn: string;
  label_bn: string;
  word_count: number;
  generated_at: IsoDateTime;
}
export type DecisionRequest =
  | { decision: "reject"; reason: string; expected_missing_version: number; expected_found_version:
number }
  | { decision: "confirm"; reason?: string; agency_name?: string;
      expected_missing_version: number; expected_found_version: number;
      reviewer_checklist: { descriptions_reviewed: true; contradictions_reviewed: true;
        identity_not_proven_by_score: true; referral_is_simulated: true } };
export interface DecisionResponse {
  candidate_status: CandidateStatus;
  missing: { ref_code: string; status: CaseStatus };
  found: { ref_code: string; status: CaseStatus };
  referral_id: Uuid | null;
  referral_state: "recorded" | null;
  referral_is_simulated: true;
}
export interface ReportCounts { open: number; matched: number; closed: number; }
export interface DistrictStats {
  code: number; name_bn: string; name_en: string; lat: number; lng: number;
  missing_reports: ReportCounts;
  found_reports: { open: number; matched: number; referral_recorded: number };
  confirmed_pairs: number;
}
export interface DashboardStatsResponse {
  generated_at: IsoDateTime;
  counting_basis: "report_rows_separated_by_kind";
  totals: { missing_reports: ReportCounts;
    found_reports: { open: number; matched: number; referral_recorded: number };
```

```
    confirmed_pairs: number };
  districts: DistrictStats[];
}
export interface CreateMissingRequest {
  draft: CaseDraft & { kind: "missing" };
  confirmed: true;
  client_submission_id: Uuid;
  draft_token?: string; // absent for manual form; paid extraction requires its own token
}
export interface CandidateDecisionView {
  candidate_id: Uuid; missing: OfficerCaseView; found: OfficerCaseView;
  score: number; components: Record<string, number | null>; explanation_version: string;
  qualified: boolean; identity_probability: false;
}
export interface DuplicateDecisionRequest {
  status: "confirmed_same" | "dismissed";
  reason: string;
  profile_evidence_reviewed: true;
}
export interface RematchResponse {
  case_id: Uuid; matching_state: MatchingState;
  // Admin gets this status-only shape. Officer reads candidates via API-008/009.
}

export interface ErrorEnvelope {
  error: {
    id: string;
    key: string;
    message_bn: string;
    message_en: string;
    retryable: boolean;
    fallback?: "manual_form";
  };
}

export type TypedCaseRecord =
  | (Case & { kind: "missing"; status: MissingStatus })
  | (Case & { kind: "found"; status: FoundStatus });
// Uuid/IsoDateTime and number ranges need runtime validation. pg numeric/bigint/Date
// values must be explicitly mapped before returning these DTOs.
```

# 7 Registry

Every ID in the suite appears in exactly one row below. "Defined in" names the file that introduces it.

## 7.1 Goals (GOAL, 9)

| ID | Label | Metric | Defined in |
|---|---|---|---|
| GOAL-001 | A parent produces a reviewed Bangla report in the first hour | Time to complete report, assistant vs manual form | 00_Gap_Check_and_Registry |
| GOAL-002 | AI extracts case fields correctly from messy Bangla | Field-level extraction accuracy | 00_Gap_Check_and_Registry |
| GOAL-003 | The right found-child record appears in the officer's shortlist | Recall@5; precision@1; precision@5 reported | 00_Gap_Check_and_Registry |

| ID | Label | Metric | Defined in |
|---|---|---|---|
| GOAL-004 | Duplicate reports of one child are detected | Duplicate-detection pairwise F1 | 00_Gap_Check_and_Registry |
| GOAL-005 | Bangla voice is usable for intake | Word error rate | 00_Gap_Check_and_Registry |
| GOAL-006 | Every demo report has an explicit recorded coordination status, visible by district | Seeded cases shown with correct status | 00_Gap_Check_and_Registry |
| GOAL-007 | The live demo does not fail | Rehearsals passing end to end | 00_Gap_Check_and_Registry |
| GOAL-008 | Humans make every match decision | Match-driven status changes with a logged officer action | 00_Gap_Check_and_Registry |
| GOAL-009 | Results are reported honestly | Metrics slide with failure cases | 00_Gap_Check_and_Registry |

## 7.2 Personas (PERSONA, 5)

| ID | Label | Profile | Defined in |
|---|---|---|---|
| PERSONA-001 | Parent/guardian ("Shirin") | Dhaka mother, 30s; son, 13, did not return from madrasa; Bangla-first, weak formal writing, And... | 00_Gap_Check_and_Registry |
| PERSONA-002 | Verifying officer ("SI Karim") | Thana duty officer, several missing-child GDs a week; reviews matches on a desktop. | 00_Gap_Check_and_Registry |
| PERSONA-003 | Shelter/station staff ("Rumana") | NGO volunteer at a railway-station children's desk; logs found children on a phone. | 00_Gap_Check_and_Registry |
| PERSONA-004 | Child-protection caseworker | Intended future referral recipient; V2 records a simulated referral only; no recipient UI or delivery integration. | 00_Gap_Check_and_Registry |
| PERSONA-005 | Disaster-mode shelter volunteer | Logs separated children after floods/cyclones; planned extension, no measured need. | 00_Gap_Check_and_Registry |

## 7.3 User stories (STORY, 23)

| ID | Label | Priority | Defined in |
|---|---|---|---|
| STORY-001 | Report by voice | Must | D1_PRD |
| STORY-002 | Report by typed text | Must | D1_PRD |
| STORY-003 | Answer follow-up questions | Must | D1_PRD |
| STORY-004 | Review before anything is shared | Must | D1_PRD |

| ID | Label | Priority | Defined in |
|---|---|---|---|
| STORY-005 | Get a GD draft | Must | D1_PRD |
| STORY-006 | Get a shareable poster | Must | D1_PRD |
| STORY-007 | Get a 13219-ready summary | Should | D1_PRD |
| STORY-008 | Log a found child | Must | D1_PRD |
| STORY-009 | See ranked candidates | Must | D1_PRD |
| STORY-010 | Compare and confirm a match | Must | D1_PRD |
| STORY-011 | Reject a candidate | Must | D1_PRD |
| STORY-012 | Refer and close | Must | D1_PRD |
| STORY-013 | See the district picture | Must | D1_PRD |
| STORY-014 | Detect duplicate reports | Must | D1_PRD |
| STORY-015 | Import synthetic public-format fixtures | Should | D1_PRD |
| STORY-016 | Redact identifying payloads from eligible terminal cases | Must | D1_PRD |
| STORY-017 | Audit decisions | Must | D1_PRD |
| STORY-018 | Continue when AI fails | Must | D1_PRD |
| STORY-019 | Edit after submission | Should | D1_PRD |
| STORY-020 | See my case status | Should | D1_PRD |
| STORY-021 | Disaster-mode logging | Could | D1_PRD |
| STORY-022 | Measure and report results | Must | D1_PRD |
| STORY-023 | Run the demo without internet | Must | D1_PRD |

## 7.4 Functional requirements (FR, 37)

| ID | Label | Priority | Defined in |
|---|---|---|---|
| FR-001 | accept typed reports of 20–2,000 characters in Bangla, English or mixed script. | Must | D1_PRD |
| FR-002 | accept voice notes of 5–120 s and return an editable Bangla transcript. | Must | D1_PRD |
| FR-003 | normalise Bangla digits and derive explicit Asia/Dhaka event intervals anchored to the recorded intake clock; ambiguous time requires review. | Must | D1_PRD |

| ID | Label | Priority | Defined in |
|---|---|---|---|
| FR-004 | extract case fields from text, leaving unstated fields empty and never guessing. | Must | D1_PRD |
| FR-005 | ask at most four one-at-a-time follow-ups, then require accessible manual completion of unresolved mandatory fields. | Must | D1_PRD |
| FR-006 | show every extracted field as editable and require an explicit confirm. | Must | D1_PRD |
| FR-007 | keep drafts out of matching, staff/officer views and exports until confirmed. | Must | D1_PRD |
| FR-008 | generate a Bangla GD draft from a fixed template using confirmed fields only. | Must | D1_PRD |
| FR-009 | generate a reviewed no-photo poster with 13219, a separate sharing acknowledgement and optional contact inclusion. | Must | D1_PRD |
| FR-010 | generate a 13219-ready summary of ≤ 80 words. | Should | D1_PRD |
| FR-011 | let staff log a found child by form or free text and review before saving. | Must | D1_PRD |
| FR-012 | rank up to five qualified candidates nationally with evidence reasons and similarity scores; allow an empty shortlist. | Must | D1_PRD |
| FR-013 | exclude candidate pairs that fail the hard filters in BR-017. | Must | D1_PRD |
| FR-014 | list cases with pending candidates for verified officers, newest candidate first. | Must | D1_PRD |
| FR-015 | show missing and found records side by side with agree/ differ/unknown per field. | Must | D1_PRD |
| FR-016 | atomically record a verified candidate decision, current endpoint states, simulated referral recorded and safe audit; no receipt or custody claim. | Must | D1_PRD |
| FR-017 | on reject with a reason, remove the candidate and apply BR-012 and BR-019. | Must | D1_PRD |
| FR-018 | record coordination_recorded closure after confirmed/ referral-recorded state, or resolved_outside_fera from open; neither proves safe reunification. | Must | D1_PRD |
| FR-019 | show separate missing-report, found-report and confirmed-pair district counts with accessible icon, text and colour. | Must | D1_PRD |
| FR-020 | restrict candidates and matched records to verified officers per BR-024. | Must | D1_PRD |
| FR-021 | write an append-only audit entry for every action listed in STORY-017. | Must | D1_PRD |

| ID | Label | Priority | Defined in |
|---|---|---|---|
| FR-022 | flag possible duplicates while keeping every unreviewed report independently discoverable; shared phone alone never suppresses a report. | Must | D1_PRD |
| FR-023 | provide optional reviewed reversible duplicate grouping/ dismissal with profile evidence, reason and preserved original reports. | Could | D1_PRD |
| FR-024 | import invented fixture texts in public-report format with a synthetic label and source metadata; real public-child import is disabled in V2. | Should | D1_PRD |
| FR-025 | redact direct and related identifying payloads only from eligible terminal reports; retain active linked evidence. | Must | D1_PRD |
| FR-026 | open a pre-filled manual form after 2 failed extractions or an AI service error. | Must | D1_PRD |
| FR-027 | in demo-safe mode, return stored outputs for scripted demo inputs with no network. | Must | D1_PRD |
| FR-028 | report frozen held-out metrics with explicit denominators, abstention/coverage, cache exclusion and failure examples. | Must | D1_PRD |
| FR-029 | let parents edit a case after submission per BR-025, keeping all versions. | Should | D1_PRD |
| FR-030 | show parents their own case status and authorised edit/ output availability without candidate or other-party evidence. | Should | D1_PRD |
| FR-031 | idempotently return one case per principal and submission UUID; identical retry returns it, a changed payload conflicts, a new UUID remains a distinct report. | Must | D1_PRD |
| FR-032 | load synthetic seed records and label them "সিনথেটিক" (synthetic) on every screen. | Must | D1_PRD |
| FR-033 | default parent screens to Bangla with an English toggle on every screen. | Must | D1_PRD |
| FR-034 | block generated or displayed text containing alert wording or "runaway" terms. | Must | D1_PRD |
| FR-035 | treat officer accounts as verified only when an admin marks them so. [MOCK: seeded accounts] | Must | D1_PRD |
| FR-036 | provide a disaster-mode logging screen labelled as a planned extension. | Could | D1_PRD |
| FR-037 | sign in staff, officers and admins with a role; parents need no account. | Must | D1_PRD |

## 7.5 Non-functional requirements (NFR, 17)

| ID | Label | Requirement | Defined in |
|---|---|---|---|
| NFR-001 | Performance | Parent intake screen usable on a mid-range Android over 4G | D1_PRD |
| NFR-002 | Performance | Text → draft | D1_PRD |
| NFR-003 | Performance | 120 s voice note → transcript | D1_PRD |
| NFR-004 | Performance | Matching for one case vs 500 records | D1_PRD |
| NFR-005 | Performance | District map with 1,000 cases | D1_PRD |
| NFR-006 | Performance | Concurrent users (A-020) | D1_PRD |
| NFR-007 | Reliability | Scripted demo journey with network off | D1_PRD |
| NFR-008 | Reliability | Draft survives network loss; fallback form after AI failure | D1_PRD |
| NFR-009 | Security | Encrypted transport; no secrets reach the browser; rate limits 10/min/user, 30/min/IP on AI act... | D1_PRD |
| NFR-010 | Privacy | No payload in logs/analytics; synthetic audio/text reaches declared providers before submit with notice; endpoint terms pending review | D1_PRD |
| NFR-011 | Accessibility | WCAG 2.2 AA; tap targets ≥ 44 × 44 px; text contrast ≥ 4.5:1 | D1_PRD |
| NFR-012 | Usability | Parent stories completed unaided; any field corrected in 1 tap | D1_PRD |
| NFR-013 | Compatibility | Android browser at 360 px width; current desktop browsers; Bangla renders with no missing glyphs | D1_PRD |
| NFR-014 | Maintainability | AI provider switched by configuration only; setup from README | D1_PRD |
| NFR-015 | Scalability | Register supports all 64 districts and 30,000 cases/year (74/day × 365 ≈ 27,000) without data-m... | D1_PRD |
| NFR-016 | Cost | Total external AI spend from 9 Oct to 24 Oct | D1_PRD |
| NFR-017 | Auditability | Audit entries immutable through the product | D1_PRD |

## 7.6 Business rules (BR, 26)

| ID | Label | Linked FRs (safeguarding rules) | Defined in |
|---|---|---|---|
| **BR-001** | AI retrieves/ranks only. Active verified officers record candidate decisions; identity and guardian verification are external pilot gates. | FR-016, FR-017, FR-020 | D1_PRD |
| **BR-002** | No issuing, withdrawing or broadcasting alerts. Outputs are reviewed drafts; no CID integration is claimed. | FR-008, FR-009, FR-010, FR-034 | D1_PRD |
| **BR-003** | No runaway labels, inferred absence reasons, or rescue promises. | FR-004, FR-034 | D1_PRD |
| **BR-004** | No submitted child payload, matching, other-party visibility, or export before reporter review/confirmation. Providers process intake after notice; pre-submit DB holds metadata only. | FR-006, FR-007 | D1_PRD |
| **BR-005** | Confirm creates a simulated internal referral record. referred means referral recorded; no receipt, custody, return, or safe reunification is inferred. | FR-016, FR-018 | D1_PRD |
| **BR-006** | No face search, photos, identity documents, exact coordinates, or physical return action in this release. | FR-009, FR-011, FR-020 | D1_PRD |
| **BR-007** | Candidates and other-party matched evidence are officer-only. Parent gets own scoped editable payload/status; staff gets own found payload/status. Admin gets aggregate/redacted metadata only. | FR-020, FR-030, FR-035 | D1_PRD |
| **BR-008** | Only specified report fields are collected. Evidence quotes refer to reviewed source text; deterministic derivations retain source spans; unknowns remain unknown. | FR-004, FR-011 | D1_PRD |
| **BR-009** | Admin redacts identifying payloads only after closure; found evidence is retained while linked missing case is unclosed. Retained IDs/date/district are pseudonymous, not irreversible anonymity. | **FR-025** | D1_PRD |
| **BR-010** | Decisions, state changes, edits, payload redaction and denied access have safe append-only audit events. Denials after rollback are logged separately; unknown case FK is null. | **FR-021** | D1_PRD |
| **BR-011** | No unsupported legal compliance claim. Accurate primary-law references are permitted. Real-data processing requires qualified partner/legal/provider review. | FR-025, NFR-010 | D1_PRD |

| ID | Label | Linked FRs (safeguarding rules) | Defined in |
|---|---|---|---|
| BR-012 | Missing report: Browser draft is not a persisted case. Initial persisted state is submitted; submitted → open; open → matched_pending_review when at least one qualified candidate exists; matched_pending_review → open when none remain; matched_pending_review... | | D1_PRD |
| BR-013 | Found report: Initial state open; open ↔ matched_pending_review; matched_pending_review → referred only in the confirmation transaction. The UI says “রেফারেল নথিভুক্ত (সিমুলেশন)” / “Referral recorded (simulation)”. closed_at on referred is the coordination ... | | D1_PRD |
| BR-014 | Counts: Missing Open = submitted/open; Missing Matched = matched_pending_review/confirmed; Missing Closed = closed. Found Open = open; Found Matched = matched_pending_review; Found Referral recorded = referred. Confirmed pairs are counted once from confirme... | | D1_PRD |
| BR-015 | Required submission fields: Missing: age or complete age range, sex selection (unknown allowed), district, non-empty last-seen place, event interval, and reporter phone. Found: description, district, place, event interval, staff contact; age may be unknown.... | | D1_PRD |
| BR-016 | Field rules: Missing ages 0–17. Exact age yields age_min = age_max = age_years; a range has both endpoints and no conflicting exact age. Found estimated ranges may extend to 25 to represent uncertainty; wholly adult profiles are not automatically treated as... | | D1_PRD |
| BR-017 | Eligibility: Opposite kinds; sex equal or unknown; age ranges overlap within ±2 years when both known, otherwise unknown is allowed with less evidence; found upper time ≥ missing lower time; live open/matched versions; neither record payload-redacted/confir... | | D1_PRD |
| BR-018 | Ranking: Retrieve eligible records nationally using exact filtered search for the ~200-record baseline. District/adjacency is a soft bonus, never a gate after five weak local results. L2-normalised embeddings share a pinned space/revision/prefix policy. Use... | | D1_PRD |
| BR-019 | Rejections: A rejected pair is not re-suggested at the same report versions. Editing creates new versions and invalidates old pending candidates. Display prior rejection history to the officer. | | D1_PRD |
| BR-020 | One-to-one by report: At most one confirmed candidate per endpoint. The transaction locks endpoints in sorted UUID order, rechecks versions/authorisation, confirms one pair, voids other pending pairs, and reconciles each affected report's queue state. Dupli... | | D1_PRD |

| ID | Label | Linked FRs (safeguarding rules) | Defined in |
|---|---|---|---|
| BR-021 | Duplicates: Shared phone creates only a weak flagged cue and cannot independently yield confirmed_same or suppress a report. Compatible profiles may create a possible-duplicate cue using dev-calibrated similarity, district and time evidence. Human review ma... | | D1_PRD |
| BR-022 | Outputs: Fixed templates use only reporter-reviewed fields plus fixed instructions and 13219. Poster export has a separate preview/share acknowledgement and contact inclusion choice. Downloads cannot be remotely recalled; no photo does not mean no sensitive... | | D1_PRD |
| BR-023 | Limits: Audio 5–120 s and ≤2 MB; codec/bitrate must satisfy both, and duration is verified server-side. Text 20–2,000 characters; at most four conversational follow-ups, then manual completion. At most six dispatched paid transcribe/ extract calls per draft,... | | D1_PRD |
| BR-024 | Permissions: Parent: own missing creation/read/edit/output only. Staff/unverified officer: own found creation/read and found-only intake. Verified officer: candidate review/decision/ closure, operational case list, aggregate dashboard and safe audit. Admin: ... | | D1_PRD |
| BR-025 | Edits: Owner edits only submitted/open/ matched_pending_review with expected_version. Increment version, snapshot the old reviewed payload, void pending pairs, update affected states, and queue bounded reliable rematching. Terminal edits are rejected; payloa... | | D1_PRD |
| BR-026 | Synthetic release: Every demo/seed/import record is synthetic, with a visible label; real public child reports are not imported in V2. Model-quality evaluation disables fixtures and caches. Mock authority, cached outputs, compressed video time, and lack of ... | | D1_PRD |

## 7.7 Assumptions (A, 40)

| ID | Label | Defined in |
|---|---|---|
| A-001 | OQ-001 – OQ-007 unanswered; Turn 0 defaults apply: voice and poster Must, 13219 Should; 7-state... | D1_PRD |
| A-002 | Numeric targets are provisional and revised after the first evaluation run on 11 Oct. | D1_PRD |
| A-003 | Personas are fictional, built from published reporting. | D1_PRD |
| A-004 | No real police, CID or MUN Alert data is available before the finale. | D1_PRD |

| ID | Label | Defined in |
|---|---|---|
| A-005 | Parents use a smartphone browser with a microphone; feature phones are out of scope. | D1_PRD |
| A-006 | Officer verification is a flag set by an admin on seeded accounts [MOCK]. | D1_PRD |
| A-007 | A referral is a recorded entry with an agency name; no external integration. | D1_PRD |
| A-008 | The GD draft uses a generic Bangla template; the official online GD field list is UNKNOWN (OQ-0… | D1_PRD |
| A-009 | The parent reads the 13219 summary on their own call; Fera never contacts 13219. | D1_PRD |
| A-010 | Usability testers are adult volunteers role-playing a parent, not real affected parents (OQ-019… | D1_PRD |
| A-011 | The manual-form baseline is the same testers filling a plain form with the same fields. | D1_PRD |
| A-012 | The speech test set is 20 scripted Bangla prompts recorded by ≥ 3 speakers. | D1_PRD |
| A-013 | The team compiles a neighbouring-district list; if absent, search goes same district, then nati… | D1_PRD |
| A-014 | Duplicate thresholds (±1 year, 72 h) are provisional and tuned on the synthetic set. | D1_PRD |
| A-015 | A 0–100 similarity score is understandable to officers if labelled as similarity. | D1_PRD |
| A-016 | Drafts expire after 24 h. | D1_PRD |
| A-017 | Demo-safe mode covers only scripted inputs for ≥ 3 named synthetic cases. | D1_PRD |
| A-018 | Hosting and database run on free tiers; the USD 30 budget covers AI services only. | D1_PRD |
| A-019 | Public reference data gives names and centroids for all 64 districts (DEP-004). | D1_PRD |
| A-020 | 50 concurrent users is a PRD v2 target; realistic demo load is ≤ 10. | D1_PRD |
| A-021 | A structured draft shortens the time to file a GD. | D1_PRD |
| A-022 | Parents accept AI-drafted text if they can review and edit it. | D1_PRD |
| A-023 | Low-literacy parents prefer speaking to typing. | D1_PRD |
| A-024 | Public found-child notices exist in enough volume to be worth matching. | D1_PRD |

| ID | Label | Defined in |
|---|---|---|
| A-025 | Officers use a ranked queue if each candidate shows evidence side by side. | D1_PRD |
| A-026 | The candidate local model (multilingual-e5-small, 384-d) gives usable Bangla similarity; verifi… | D2_TRD_PartA |
| A-027 | The local model loads within the hosting function's size and time limits; if not, EMBED_PROVIDE… | D2_TRD_PartA |
| A-028 | Match score = 100 × (0.55 cosine + 0.15 name + 0.15 clothing + 0.10 district + 0.05 age closene… | D2_TRD_PartA |
| A-029 | Duplicate profile threshold is a development-only hypothesis, calibrated on hard negatives and frozen before held-out test; phone alone never confirms a duplicate. | D2_TRD_PartA |
| A-030 | Android Chrome MediaRecorder output (webm/opus) is accepted by the speech provider; verified 10… | D2_TRD_PartA |
| A-031 | Browsers allow microphone access only on HTTPS or localhost. The finale phone reaches the deplo… | D2_TRD_PartA |
| A-032 | District codes 1–64 are Fera-internal, not official geocodes. | D2_TRD_PartA |
| A-033 | The team builds a Bangla→Latin romanisation table for name similarity. | D2_TRD_PartA |
| A-034 | A free-tier managed Postgres with pgvector >= 0.8.1 (HNSW) is available (OQ-021). | D2_TRD_PartA |
| A-035 | Time words map to fixed hours: সকাল 08:00, দুপুর 13:00, বিকেল 16:00, সন্ধ্যা 18:30, রাত 21:00, … | D2_TRD_PartA |
| A-036 | 2 MB is enough for a 120 s webm/opus note from Android Chrome. | D2_TRD_PartA |
| A-037 | Usage volumes in §12.2 (190 audio minutes, 1,100 extraction calls, ~1,200 input + 400 output to… | D2_TRD_PartB |
| A-038 | Four-times output weighting is a budget assumption; verify actual input/output rates at the 11 Oct contract gate. | D2_TRD_PartB |
| A-039 | 200 total reports: 80 missing, 80 found, 40 additional decoys; 60 canonical true pairs, 20 no-match missing queries, 25 duplicate links within the total. | D2_TRD_PartB |
| A-040 | Actual availability and completed work must be inventoried on 11 Oct. A fresh 11–14 Oct start at 10 h/person/day gives at most 120 team-hours; this is an assumption, not verified capacity. | D6_Implementation_Plan |

## 7.8 Dependencies (DEP, 11)

| ID | Label | Owner | Defined in |
|---|---|---|---|
| DEP-001 | Speech-to-text service with Bangla support | Obidit | D1_PRD |
| DEP-002 | At least two interchangeable language-model services | Obidit | D1_PRD |
| DEP-003 | Text-similarity capability that also works offline for scripted inputs | Masnun | D1_PRD |
| DEP-004 | District reference data: 64 districts, Bangla/ English names, centroids | Masnun | D1_PRD |
| DEP-005 | Synthetic set, ~200 records, known pairs, duplicates, decoys | Masnun | D1_PRD |
| DEP-006 | 50 hand-labelled reports | Masnun, Obidit | D1_PRD |
| DEP-007 | Public news and alert texts (public pages only) | Masnun | D1_PRD |
| DEP-008 | Free-tier hosting and database | Sadman | D1_PRD |
| DEP-009 | Finale kit: laptop, Android phone, mobile hotspot | Sadman | D1_PRD |
| DEP-010 | Mentor or organiser answers to OQ-003 and OQ-010 | Sadman | D1_PRD |
| DEP-011 | Base map display; offline fallback without map tiles | Masnun | D1_PRD |

## 7.9 Risks (RISK, 17)

| ID | Label | Likelihood | Defined in |
|---|---|---|---|
| RISK-001 | Core journey unfinished by 14 Oct official cutoff (time pending) | M | D1_PRD |
| RISK-002 | Bangla speech recognition too poor | H | D1_PRD |
| RISK-003 | AI invents or misreads fields | M | D1_PRD |
| RISK-004 | AI service outage, quota or spend over USD 30 | M | D1_PRD |
| RISK-005 | Venue internet fails at the finale | M | D1_PRD |
| RISK-006 | Matching below targets | M | D1_PRD |
| RISK-007 | Synthetic set too easy, inflating metrics | M | D1_PRD |
| RISK-008 | Stalker or trafficker misuse | L | D1_PRD |
| RISK-009 | False match sends a child into danger | L | D1_PRD |

| ID | Label | Likelihood | Defined in |
|---|---|---|---|
| RISK-010 | Judges favour environment-themed projects | M | D1_PRD |
| RISK-011 | "Why not just MUN Alert?" | H | D1_PRD |
| RISK-012 | Overclaiming on metrics, interviews or law | M | D1_PRD |
| RISK-013 | Bangla text renders badly on low-end phones | L | D1_PRD |
| RISK-014 | Instructions hidden in imported public text steer the AI | M | D1_PRD |
| RISK-015 | Local embedding model too large or slow for serverless hosting | M | D2_TRD_PartA |
| RISK-016 | Phone microphone blocked when the app is served over plain HTTP in local mode | M | D2_TRD_PartA |
| RISK-017 | V2 Round 2 estimates total 184 h and may exceed actual remaining capacity | M | D6_Implementation_Plan |

## 7.10 Open questions (OQ, 26)

| ID | Label | Context / default / owner | Defined in |
|---|---|---|---|
| OQ-001 | Are **voice intake** and **poster + 13219 summary** Must for Round 2? | PRD v2 marks both Should (v2 FR7, FR8). PROJECT CONTEXT lists "Bangla intake → poster + GD" as ... | 00_Gap_Check_and_Registry |
| OQ-002 | Adopt the separate missing/found state machine from PROJECT CONTEXT (draft → submitted → open → matched-pending-review → confirmed → closed/referred)? | PRD v2 uses open / matched / closed / reopened and closes both cases on confirm. Changes BRs, s... | 00_Gap_Check_and_Registry |
| OQ-003 | What did mentors say in session 1 (7–8 Oct) about the environment theme and disaster mode? | PRD v2 rates "judges favour environmental projects" Medium/High risk. The answer decides whethe... | 00_Gap_Check_and_Registry |
| OQ-004 | Which accounts and API keys does the team already hold (LLM providers, OpenAI for Whisper, Postgres host, Vercel)? Any free credits? | Sets the default LLM provider, the embedding source and the USD 30 budget estimate. | 00_Gap_Check_and_Registry |
| OQ-005 | Finale demo topology: deployed app over venue Wi-Fi, or laptop-local (localhost DB, phone on laptop hotspot)? | Venue internet may fail. Local-first changes DB hosting, embeddings and the fallback design. | 00_Gap_Check_and_Registry |

| ID | Label | Context / default / owner | Defined in |
|---|---|---|---|
| OQ-006 | What exists today (9 Oct)? Repo skeleton, synthetic set, 50 labelled reports, any user conversations held? | Actual repository progress and user/ evaluation evidence are UNVERIFIED; inventory now. | 00_Gap_Check_and_Registry |
| OQ-007 | Demo auth and roles: real login or a role switcher? Keep PRD v2's extra "helper" role? | PRD v2 roles: parent, helper, staff, officer, admin. PROJECT CONTEXT: parent, officer, shelter ... | 00_Gap_Check_and_Registry |
| OQ-008 | Evidence scope | PROJECT CONTEXT says use only 5 figures. Deck and PRD v2 also cite age 0–9 split, MUN Alert "20... | 00_Gap_Check_and_Registry |
| OQ-009 | Accessibility level | PRD v2 targets WCAG 2.1 AA; D4 spec asks 2.2 AA. | 00_Gap_Check_and_Registry |
| OQ-010 | Alert authority wording | Deck slide 4 says only CID's Missing Children Cell can issue alerts, and also that a proposed C... | 00_Gap_Check_and_Registry |
| OQ-011 | Personal Data Protection Act (2026) | PRD v2 says "passed April 2026". No clauses are in any source. | 00_Gap_Check_and_Registry |
| OQ-012 | Retention period for closed cases | PRD v2: "to be set with a partner". Safeguarding requires delete/redact identifying payloads. | 00_Gap_Check_and_Registry |
| OQ-013 | LLM/STT provider data terms | PRD v2: "confirm per provider" no-training and retention terms. | 00_Gap_Check_and_Registry |
| OQ-014 | Persona count | PRD v2 has 4 personas incl. child-protection officer; D1 spec asks for 3. | 00_Gap_Check_and_Registry |
| OQ-015 | Dedupe priority | PRD v2 puts duplicate merge at Could (v2 FR10). Deck promises a duplicate-detection quality metric... | 00_Gap_Check_and_Registry |
| OQ-016 | Official online GD field list and format? | UNKNOWN; generic template (A-008) | D1_PRD |
| OQ-017 | Source for neighbouring-district adjacency? | Team-compiled (A-013) | D1_PRD |
| OQ-018 | Which child-protection agency name should a referral show? | UNKNOWN; demo shows "শিশু সুরক্ষা সেবা" (child-protection services) | D1_PRD |
| OQ-019 | Can usability tests ethically include real affected parents? | Default: no (A-010) | D1_PRD |
| OQ-020 | Should parents return to a case from another device? | Default: Won't this release | D1_PRD |

| ID | Label | Context / default / owner | Defined in |
|---|---|---|---|
| OQ-021 | Managed Postgres host | Pick on 10 Oct; must pass CREATE EXTENSION vector and HNSW index creation | D2_TRD_Part A |
| OQ-022 | Speech provider limits and price | Verify from provider docs on 10 Oct; record in Part B §12 | D2_TRD_Part A |
| OQ-023 | Hosting limits for local embeddings | 10 Oct spike; fallback A-027 | D2_TRD_Part A |
| OQ-024 | Do classes, exams or labs reduce team hours before 14 Oct? | UNKNOWN; inventory actual availability and finished work at the 11 Oct review | D6_Implementation_Plan |
| OQ-025 | Judging weights for impact, feasibility, scalability | Organisers | D7_Closing_Pack |
| OQ-026 | Is synthetic data without real public-child news acceptable for Round 2 technical evaluation? | Organisers / mentors | D7_Closing_Pack |

## 7.11 Screens (SCREEN, 16)

| ID | Label | Priority | Defined in |
|---|---|---|---|
| SCREEN-001 | Welcome and language | Must | D3_App_Flow |
| SCREEN-002 | Intake chat (voice/text) | Must | D3_App_Flow |
| SCREEN-003 | Case review | Must | D3_App_Flow |
| SCREEN-004 | Case submitted and outputs | Must | D3_App_Flow |
| SCREEN-005 | Manual case form | Must | D3_App_Flow |
| SCREEN-006 | Parent case status and edit | Should | D3_App_Flow |
| SCREEN-007 | Found-child log | Must | D3_App_Flow |
| SCREEN-008 | Sign-in | Must | D3_App_Flow |
| SCREEN-009 | Review queue | Must | D3_App_Flow |
| SCREEN-010 | Match review (side by side) | Must | D3_App_Flow |
| SCREEN-011 | District dashboard | Must | D3_App_Flow |
| SCREEN-012 | Case detail and audit trail | Must | D3_App_Flow |
| SCREEN-013 | Admin data tools | Must | D3_App_Flow |
| SCREEN-014 | Duplicate review | Could | D3_App_Flow |

| ID | Label | Priority | Defined in |
|---|---|---|---|
| SCREEN-015 | Disaster mode | Could | D3_App_Flow |
| SCREEN-016 | Not found / not allowed | Must | D3_App_Flow |

## 7.12 Components (server COMP-001–021; UI COMP-022–050) (COMP, 50)

| ID | Label | Detail | Defined in |
|---|---|---|---|
| COMP-001 | Parent client | Screens SCREEN-001–006; draft in localStorage | D2_TRD_PartA |
| COMP-002 | Staff/officer/admin client | SCREEN-007–016 | D2_TRD_PartA |
| COMP-003 | Voice recorder | Capture 5–120 s via MediaRecorder; show timer | D2_TRD_PartA |
| COMP-004 | API layer | Route handlers; Zod; auth guard; rate limit; request ID | D2_TRD_PartA |
| COMP-005 | Auth and session | Auth.js credentials; parent cookie | D2_TRD_PartA |
| COMP-006 | Speech adapter | Transcribe; script check; cache lookup | D2_TRD_PartA |
| COMP-007 | LLM adapter | Extraction via contract §2.1; retry; fallback | D2_TRD_PartA |
| COMP-008 | Normaliser | Bangla numerals → ASCII; dates/time words → BST ISO; phone; district lookup (bn/en/ alias); roma... | D2_TRD_PartA |
| COMP-009 | Intake orchestrator | Merge extraction + answers; evidence check; required-field check; templated question; AI call c... | D2_TRD_PartA |
| COMP-010 | Output generator | GD, poster text, 13219 summary from templates | D2_TRD_PartA |
| COMP-011 | Embedding service | Build case description string; embed | D2_TRD_PartA |
| COMP-012 | Matching engine | Hard filters (SQL), cosine (pgvector), bonuses, top 5, template reason | D2_TRD_PartA |
| COMP-013 | Duplicate detector | Apply D1 BR-021; write duplicate_links | D2_TRD_PartA |
| COMP-014 | Case service + state guard | Create, edit (versions), transitions, referral, close, redact identifying payloads | D2_TRD_PartA |

| ID | Label | Detail | Defined in |
|---|---|---|---|
| COMP-015 | Audit logger | Append audit_log rows inside the same transaction | D2_TRD_PartA |
| COMP-016 | Wording guard | Block alert wording and "runaway" terms in Fera-generated text and labels | D2_TRD_PartA |
| COMP-017 | Demo-safe cache | Fixtures for scripted inputs: transcript, extraction, embeddings | D2_TRD_PartA |
| COMP-018 | Dashboard and map | Aggregate counts; Leaflet map; table view | D2_TRD_PartA |
| COMP-019 | Admin tools | Seed loader (CLI), public import, redact identifying payloadsr, verify officer [MOCK] | D2_TRD_PartA |
| COMP-020 | Evaluation harness | Compute 5 metrics; write report | D2_TRD_PartA |
| COMP-021 | Rate limiter | Fixed-window counters in Postgres | D2_TRD_PartA |
| COMP-022 | App header with language toggle | All | D4_Design_Brief |
| COMP-023 | Synthetic badge "সিনথেটিক" | Any seeded record | D4_Design_Brief |
| COMP-024 | Chat bubble (assistant / user) | SCREEN-002 | D4_Design_Brief |
| COMP-025 | Voice record button with timer | SCREEN-002, SCREEN-007 | D4_Design_Brief |
| COMP-026 | Transcript editor | SCREEN-002 | D4_Design_Brief |
| COMP-027 | Question prompt with quick-reply chips | SCREEN-002 | D4_Design_Brief |
| COMP-028 | Editable field card | SCREEN-003, SCREEN-007 | D4_Design_Brief |
| COMP-029 | Confirm bar (checkbox + submit) | SCREEN-003, SCREEN-007 | D4_Design_Brief |
| COMP-030 | Output card with copy | SCREEN-004 | D4_Design_Brief |
| COMP-031 | Poster preview + PNG download | SCREEN-004 | D4_Design_Brief |
| COMP-032 | Manual case form | SCREEN-005 | D4_Design_Brief |
| COMP-033 | Status badge (icon + text + colour) | SCREEN-004, -006, -009, -011, -012 | D4_Design_Brief |

| ID | Label | Detail | Defined in |
|---|---|---|---|
| COMP-034 | Found-child form | SCREEN-007, SCREEN-015 | D4_Design_Brief |
| COMP-035 | Review queue list | SCREEN-009 | D4_Design_Brief |
| COMP-036 | Candidate row (rank, score, reason) | SCREEN-009 | D4_Design_Brief |
| COMP-037 | Side-by-side evidence table | SCREEN-010, SCREEN-014 | D4_Design_Brief |
| COMP-038 | Decision panel (confirm / reject + reason) | SCREEN-010 | D4_Design_Brief |
| COMP-039 | Close-case dialog | SCREEN-012 | D4_Design_Brief |
| COMP-040 | District map | SCREEN-011 | D4_Design_Brief |
| COMP-041 | District table | SCREEN-011 | D4_Design_Brief |
| COMP-042 | Audit timeline | SCREEN-012 | D4_Design_Brief |
| COMP-043 | Admin action panel | SCREEN-013 | D4_Design_Brief |
| COMP-044 | Inline alert / toast | All | D4_Design_Brief |
| COMP-045 | Empty state | SCREEN-006, -009, -013, -016 | D4_Design_Brief |
| COMP-046 | Loading state with 10 s message | SCREEN-002, -004, -007 | D4_Design_Brief |
| COMP-047 | Sign-in form | SCREEN-008 | D4_Design_Brief |
| COMP-048 | Cached badge "ক্যাশ" | Any AI output from demo cache | D4_Design_Brief |
| COMP-049 | Duplicate banner "সম্ভাব্য একই শিশু" | SCREEN-010, SCREEN-012 | D4_Design_Brief |
| COMP-050 | Consent notice | SCREEN-001 | D4_Design_Brief |

## 7.13 API endpoints (API, 23)

| ID | Label | Priority | Defined in |
|---|---|---|---|
| API-001 | POST /api/intake/transcribe | Must | D2_TRD_PartA |
| API-002 | POST /api/intake/extract | Must | D2_TRD_PartA |
| API-003 | POST /api/cases | Must | D2_TRD_PartA |
| API-004 | GET /api/cases/:id | Must | D2_TRD_PartA |
| API-005 | PATCH /api/cases/:id | Should | D2_TRD_PartA |

| ID | Label | Priority | Defined in |
|---|---|---|---|
| API-006 | POST /api/cases/:id/outputs | Must (gd, poster) / Should (helpline) | D2_TRD_PartA |
| API-007 | POST /api/found-cases | Must | D2_TRD_PartA |
| API-008 | GET /api/review-queue | Must | D2_TRD_PartA |
| API-009 | GET /api/candidates/:id | Must | D2_TRD_PartA |
| API-010 | POST /api/candidates/:id/decision | Must | D2_TRD_PartA |
| API-011 | POST /api/cases/:id/close | Must | D2_TRD_PartA |
| API-012 | GET /api/dashboard/stats | Must | D2_TRD_PartA |
| API-013 | GET /api/cases/:id/audit | Must | D2_TRD_PartA |
| API-014 | POST /api/match/:caseId | Must | D2_TRD_PartA |
| API-015 | POST /api/admin/import | Should | D2_TRD_PartA |
| API-016 | POST /api/admin/anonymise | Must | D2_TRD_PartA |
| API-017 | POST /api/admin/users/:id/verify | Must [MOCK] | D2_TRD_PartA |
| API-018 | GET /api/my-case | Could | D2_TRD_PartA |
| API-019 | GET /api/health | Must | D2_TRD_PartA |
| API-020 | /api/auth/* | Must | D2_TRD_PartA |
| API-021 | GET /api/admin/users | Must | D2_TRD_PartA |
| API-022 | GET /api/cases (officer district list) | Must | D2_TRD_PartA |
| API-023 | POST /api/duplicates/:id/decision | Could | D2_TRD_PartA |

## 7.14 Entities (ENTITY, 17)

| ID | Label | Table | Defined in |
|---|---|---|---|
| ENTITY-001 | User | users | D2_TRD_PartA |
| ENTITY-002 | ParentSession | parent_sessions | D2_TRD_PartA |
| ENTITY-003 | Case | cases | D2_TRD_PartA |
| ENTITY-004 | CaseVersion | case_versions | D2_TRD_PartA |

| ID | Label | Table | Defined in |
|---|---|---|---|
| ENTITY-005 | MatchCandidate | match_candidates | D2_TRD_PartA |
| ENTITY-006 | ReviewDecision | review_decisions | D2_TRD_PartA |
| ENTITY-007 | Referral | referrals | D2_TRD_PartA |
| ENTITY-008 | DuplicateLink | duplicate_links | D2_TRD_PartA |
| ENTITY-009 | AuditEntry | audit_log | D2_TRD_PartA |
| ENTITY-010 | District | districts | D2_TRD_PartA |
| ENTITY-011 | DistrictNeighbour | district_neighbours | D2_TRD_PartA |
| ENTITY-012 | GeneratedOutput | generated_outputs | D2_TRD_PartA |
| ENTITY-013 | AiCall | ai_calls | D2_TRD_PartA |
| ENTITY-014 | RateLimitBucket | rate_limits | D2_TRD_PartA |
| ENTITY-015 | IntakeDraft | `intake_drafts` | D2_TRD_PartA |
| ENTITY-016 | AiBudget | `ai_budget` | D2_TRD_PartA |
| ENTITY-017 | AiReservation | `ai_reservations` | D2_TRD_PartA |

## 7.15 Decisions (DECISION, 17)

| ID | Label | Section | Defined in |
|---|---|---|---|
| DECISION-001 | Single Next.js App Router app (UI + API), TypeScript strict | §2 | D2_TRD_PartA |
| DECISION-002 | PostgreSQL 16 + pgvector; raw SQL via `pg`; `schema_v2.sql` is the migration | §2, §7 | D2_TRD_PartA |
| DECISION-003 | Auth.js credentials for staff/officer/admin; signed anonymous cookie for parents | §9 | D2_TRD_PartA |
| DECISION-004 | LLM used for extraction only, behind a fixed JSON contract, temp 0, 1 retry, 1 fallback provider | §2.1 | D2_TRD_PartA |
| DECISION-005 | Speech via `SpeechProvider`; Whisper API default; cache for scripted notes | §2.2 | D2_TRD_PartA |
| DECISION-006 | Local embeddings (Transformers.js, 384-d) by default; API embeddings as switchable fallback | §2.3 | D2_TRD_PartA |
| DECISION-007 | Follow-up questions, GD, poster, 13219 summary and match reasons from templates, not LLM | §4 | D2_TRD_PartA |
| DECISION-008 | Matching = SQL hard filters + pgvector cosine + rule bonuses | §4 | D2_TRD_PartA |

| ID | Label | Section | Defined in |
|---|---|---|---|
| DECISION-009 | Leaflet with OSM tiles online; tile-less markers plus table offline | §2 | D2_TRD_Part A |
| DECISION-010 | Vercel for Round 2; laptop-local copy for the finale; phone uses its own mobile data | §3.6 | D2_TRD_Part A |
| DECISION-011 | Rate limits stored in Postgres (serverless-safe) | §4 | D2_TRD_Part A |
| DECISION-012 | No optimistic updates for any state-changing action | §8 | D2_TRD_Part A |
| DECISION-013 | No per-case coordinates; district centroid only | §6 | D2_TRD_Part A |
| DECISION-014 | Draft held in the browser; nothing persisted before confirm | §3.4, §8 | D2_TRD_Part A |
| DECISION-015 | Poster PNG rendered in the browser via canvas | §2 | D2_TRD_Part A |
| DECISION-016 | Dev/test split for matching evaluation | | D2_TRD_Part B |
| DECISION-017 | Hard AI spend cap in app | | D2_TRD_Part B |

## 7.16 Error codes (E, 24)

| ID | Label | HTTP | Defined in |
|---|---|---|---|
| E-001 | VALIDATION_FAILED | 400 | D2_TRD_PartA |
| E-002 | UNAUTHENTICATED | 401 | D2_TRD_PartA |
| E-003 | FORBIDDEN_ACTION | 403 | D2_TRD_PartA |
| E-004 | NOT_FOUND | 404 | D2_TRD_PartA |
| E-005 | ILLEGAL_TRANSITION | 409 | D2_TRD_PartA |
| E-006 | SUBMISSION_PAYLOAD_CONFLICT | 409 | D2_TRD_PartA |
| E-007 | RATE_LIMITED | 429 | D2_TRD_PartA |
| E-008 | AUDIO_TOO_LARGE | 413 | D2_TRD_PartA |
| E-009 | AUDIO_LENGTH_OUT_OF_RANGE | 422 | D2_TRD_PartA |
| E-010 | SPEECH_UNINTELLIGIBLE | 422 | D2_TRD_PartA |
| E-011 | SPEECH_UNAVAILABLE | 503 | D2_TRD_PartA |

| ID | Label | HTTP | Defined in |
|---|---|---|---|
| E-012 | LLM_INVALID_OUTPUT | 422 | D2_TRD_PartA |
| E-013 | LLM_UNAVAILABLE | 503 | D2_TRD_PartA |
| E-014 | AI_TIMEOUT | 504 | D2_TRD_PartA |
| E-015 | UNSUPPORTED_LANGUAGE | 422 | D2_TRD_PartA |
| E-016 | OUTPUT_BLOCKED | 422 | D2_TRD_PartA |
| E-017 | REFERRAL_REQUIRED | 409 | D2_TRD_PartA |
| E-018 | EDIT_LOCKED | 409 | D2_TRD_PartA |
| E-019 | SESSION_EXPIRED | 401 | D2_TRD_PartA |
| E-020 | OUTPUT_REQUEST_LIMIT_RESERVED | 429 | D2_TRD_PartA |
| E-021 | INTERNAL | 500 | D2_TRD_PartA |
| E-022 | MIC_UNAVAILABLE | — | D2_TRD_PartA |
| E-023 | AI_BUDGET_CAP | 429 | D2_TRD_PartA |
| E-024 | NETWORK_OFFLINE | client only | D2_TRD_PartB |

## 7.17 Test cases (TC, 66)

| ID | Label | Defined in |
|---|---|---|
| TC-001 | Voice intake | D6_Implementation_Plan |
| TC-002 | Text intake, mixed script, Bangla numerals | D6_Implementation_Plan |
| TC-003 | Follow-up questions | D6_Implementation_Plan |
| TC-004 | Review before share | D6_Implementation_Plan |
| TC-005 | GD draft fields, no invented facts | D6_Implementation_Plan |
| TC-006 | Poster, no photo | D6_Implementation_Plan |
| TC-007 | 13219 summary ≤ 80 words | D6_Implementation_Plan |
| TC-008 | Found-child log | D6_Implementation_Plan |
| TC-009 | Top-5 candidates, hard filters | D6_Implementation_Plan |
| TC-010 | Confirm → referral → close | D6_Implementation_Plan |
| TC-011 | Reject → open, not re-suggested | D6_Implementation_Plan |

| ID | Label | Defined in |
|---|---|---|
| TC-012 | Map counts equal register | D6_Implementation_Plan |
| TC-013 | Duplicate flag and link | D6_Implementation_Plan |
| TC-014 | Public import with source URL | D6_Implementation_Plan |
| TC-015 | Related-payload redaction | D6_Implementation_Plan |
| TC-016 | Audit completeness | D6_Implementation_Plan |
| TC-017 | Manual form fallback | D6_Implementation_Plan |
| TC-018 | Edit after submission | D6_Implementation_Plan |
| TC-019 | Parent status view hides match data | D6_Implementation_Plan |
| TC-020 | Evaluation report | D6_Implementation_Plan |
| TC-021 | Offline scripted demo | D6_Implementation_Plan |
| TC-022 | Unauthorised access to matched record | D6_Implementation_Plan |
| TC-023 | Non-officer cannot confirm | D6_Implementation_Plan |
| TC-024 | No alert or "runaway" wording | D6_Implementation_Plan |
| TC-025 | Outputs labelled as drafts | D6_Implementation_Plan |
| TC-026 | No photo upload or face search | D6_Implementation_Plan |
| TC-027 | Data minimisation | D6_Implementation_Plan |
| TC-028 | No personal data in logs | D6_Implementation_Plan |
| TC-029 | One-to-one confirm lock | D6_Implementation_Plan |
| TC-030 | Field validation | D6_Implementation_Plan |
| TC-031 | Scoped submission UUID idempotency | D6_Implementation_Plan |
| TC-032 | Illegal state transitions rejected | D6_Implementation_Plan |
| TC-033 | Accessibility audit | D6_Implementation_Plan |
| TC-034 | Performance targets | D6_Implementation_Plan |
| TC-035 | Disaster-mode screen | D6_Implementation_Plan |
| TC-036 | AI spend tracking | D6_Implementation_Plan |
| TC-037 | No legal compliance claims | D6_Implementation_Plan |
| TC-038 | Shared-phone distinct children | D6_Implementation_Plan |

| ID | Label | Defined in |
|---|---|---|
| TC-039 | Unknown found age | D6_Implementation_Plan |
| TC-040 | Overlapping time uncertainty | D6_Implementation_Plan |
| TC-041 | No plausible candidate | D6_Implementation_Plan |
| TC-042 | Concurrent submission UUID | D6_Implementation_Plan |
| TC-043 | Draft replay/concurrency | D6_Implementation_Plan |
| TC-044 | Admin rematch projection | D6_Implementation_Plan |
| TC-045 | Owner editable read | D6_Implementation_Plan |
| TC-046 | Stale candidate decision | D6_Implementation_Plan |
| TC-047 | Wrong referral endpoints | D6_Implementation_Plan |
| TC-048 | Reject without reason | D6_Implementation_Plan |
| TC-049 | Terminal report edit | D6_Implementation_Plan |
| TC-050 | Related note redaction | D6_Implementation_Plan |
| TC-051 | Audit truncate/privilege | D6_Implementation_Plan |
| TC-052 | Unknown-resource denial | D6_Implementation_Plan |
| TC-053 | Last competitor invalidated | D6_Implementation_Plan |
| TC-054 | Matching fails after submit | D6_Implementation_Plan |
| TC-055 | One pair dashboard lifecycle | D6_Implementation_Plan |
| TC-056 | Evaluation fixture contamination | D6_Implementation_Plan |
| TC-057 | Single-DB offline scenario | D6_Implementation_Plan |
| TC-058 | Reflow/keyboard/Bangla | D6_Implementation_Plan |
| TC-059 | Embedding space switch | D6_Implementation_Plan |
| TC-060 | Provider notice/metadata | D6_Implementation_Plan |
| TC-061 | Encoded audio limits | D6_Implementation_Plan |
| TC-062 | PostgreSQL16/runtime host | D6_Implementation_Plan |
| TC-063 | Correction/reopen pilot gate | D6_Implementation_Plan |
| TC-064 | Manifest/group split | D6_Implementation_Plan |
| TC-065 | Concurrent AI budget | D6_Implementation_Plan |
| TC-066 | Legal claims/source accuracy | D6_Implementation_Plan |

## 7.18 Flow actions (ACT, 33)

| ID | Label | Role | Defined in |
|---|---|---|---|
| ACT-001 | Tap Start | Parent | D3_App_Flow |
| ACT-002 | Record voice note | Parent, staff | D3_App_Flow |
| ACT-003 | Type report or description | Parent, staff | D3_App_Flow |
| ACT-004 | Edit transcript | Parent | D3_App_Flow |
| ACT-005 | Send / next | Parent | D3_App_Flow |
| ACT-006 | Answer follow-up question | Parent | D3_App_Flow |
| ACT-007 | Edit a field | Parent, staff | D3_App_Flow |
| ACT-008 | Check and submit | Parent | D3_App_Flow |
| ACT-009 | Request GD draft | Parent | D3_App_Flow |
| ACT-010 | Request poster | Parent | D3_App_Flow |
| ACT-011 | Request 13219 summary | Parent | D3_App_Flow |
| ACT-012 | Copy text | Parent | D3_App_Flow |
| ACT-013 | Download poster PNG | Parent | D3_App_Flow |
| ACT-014 | Open manual form | Parent | D3_App_Flow |
| ACT-015 | Discard / cancel | Parent, staff | D3_App_Flow |
| ACT-016 | View case status | Parent | D3_App_Flow |
| ACT-017 | Correct details after submit | Parent | D3_App_Flow |
| ACT-018 | Sign in | Staff, officer, admin | D3_App_Flow |
| ACT-019 | Save found child | Staff, officer | D3_App_Flow |
| ACT-020 | Open candidate | Officer | D3_App_Flow |
| ACT-021 | Confirm match | Officer | D3_App_Flow |
| ACT-022 | Reject match with reason | Officer | D3_App_Flow |
| ACT-023 | Close case with outcome | Officer | D3_App_Flow |
| ACT-024 | Use map or district table | Officer, admin | D3_App_Flow |
| ACT-025 | View audit trail | Officer, admin | D3_App_Flow |
| ACT-026 | Import invented fixture texts | Admin | D3_App_Flow |

| ID | Label | Role | Defined in |
|---|---|---|---|
| ACT-027 | Redact identifying payloads from eligible terminal cases | Admin | D3_App_Flow |
| ACT-028 | Verify officer [MOCK] | Admin | D3_App_Flow |
| ACT-029 | Sign out | Staff, officer, admin | D3_App_Flow |
| ACT-030 | Switch language | All | D3_App_Flow |
| ACT-031 | Retry / continue draft | Parent, staff | D3_App_Flow |
| ACT-032 | Mark duplicate same/ different | Officer | D3_App_Flow |
| ACT-033 | Log separated child (disaster mode) | Staff | D3_App_Flow |

## 7.19 Flow decisions (D, 16)

| ID | Label | Defined in |
|---|---|---|
| D-001 | Microphone available and permitted? | D3_App_Flow |
| D-002 | Audio length 5–120 s and ≤ 2 MB? | D3_App_Flow |
| D-003 | Transcript intelligible and Bangla/ English? | D3_App_Flow |
| D-004 | Required fields complete? | D3_App_Flow |
| D-005 | AI calls used < 6? | D3_App_Flow |
| D-006 | Extraction valid within 2 attempts? | D3_App_Flow |
| D-007 | Submission UUID retry or conflicting payload? | D3_App_Flow |
| D-008 | Role officer and verified? | D3_App_Flow |
| D-009 | Pending candidates exist? | D3_App_Flow |
| D-010 | Officer confirms or rejects? | D3_App_Flow |
| D-011 | Other pending candidates remain? | D3_App_Flow |
| D-012 | Referral exists? | D3_App_Flow |
| D-013 | Case editable (BR-025)? | D3_App_Flow |
| D-014 | Device online? | D3_App_Flow |
| D-015 | Session valid? | D3_App_Flow |
| D-016 | Admin typed confirmation correct? | D3_App_Flow |

## 7.20 Tasks (TASK, 64)

| ID | Label | Priority | Defined in |
|---|---|---|---|
| TASK-001 | Repo: Next.js App Router, TS strict, Tailwind, ESLint, CI (lint, typecheck, test) | Must | D6_Implementation_Plan |
| TASK-002 | Pick DB host (OQ-021); apply `schema_v2.sql`; local Docker DB | Must | D6_Implementation_Plan |
| TASK-003 | Provider keys, `.env.example`, record real prices in `docs/costs.md` (OQ-004, OQ-022) | Must | D6_Implementation_Plan |
| TASK-004 | Spike: local embedding model on Vercel (OQ-023, RISK-015) | Must | D6_Implementation_Plan |
| TASK-005 | Spike: Android MediaRecorder → STT (A-030) | Must | D6_Implementation_Plan |
| TASK-006 | Auth.js credentials, parent cookie, `requireRole`, seeded users | Must | D6_Implementation_Plan |
| TASK-007 | API layer: error envelope E-001 – E-024, Zod helpers, request ID, rate limiter (COMP-021) | Must | D6_Implementation_Plan |
| TASK-008 | Layout shell, `tokens_v2.css`, Noto Sans Bengali, language toggle (COMP-022) | Must | D6_Implementation_Plan |
| TASK-009 | District data: 64 rows bn/en/centroids + neighbours (DEP-004, A-013) | Must | D6_Implementation_Plan |
| TASK-010 | Synthetic set generator: ~200 records, ground truth, duplicates, decoys (A-039) | Must | D6_Implementation_Plan |
| TASK-011 | 50 hand-labelled reports | Must | D6_Implementation_Plan |
| TASK-012 | Normaliser: Bangla digits, time words, phone, district aliases, romaniser (COMP-008) | Must | D6_Implementation_Plan |
| TASK-013 | LLM adapter: 2 providers, Zod contract, evidence check, retry, fallback (COMP-007) | Must | D6_Implementation_Plan |
| TASK-014 | Speech adapter + API-001 (COMP-006) | Must | D6_Implementation_Plan |
| TASK-015 | Case service, state guard, audit, versions (COMP-014, COMP-015) | Must | D6_Implementation_Plan |
| TASK-016 | Embedding service + `npm run reembed` (COMP-011) | Must | D6_Implementation_Plan |
| TASK-017 | Matching engine: filters, cosine, bonuses, template reasons (COMP-012) | Must | D6_Implementation_Plan |
| TASK-018 | Duplicate detector (COMP-013) | Must | D6_Implementation_Plan |

| ID | Label | Priority | Defined in |
|---|---|---|---|
| TASK-019 | Seed loader with synthetic labels and 3 demo cases | Must | D6_Implementation_Plan |
| TASK-020 | Intake orchestrator + API-002 + question templates (COMP-009) | Must | D6_Implementation_Plan |
| TASK-021 | SCREEN-001/002 intake chat + recorder (COMP-024 – COMP-027) | Must | D6_Implementation_Plan |
| TASK-022 | SCREEN-003 review + API-003 + draft persistence | Must | D6_Implementation_Plan |
| TASK-023 | Output templates GD/poster + wording guard + API-006 (COMP-010, COMP-016) | Must | D6_Implementation_Plan |
| TASK-024 | SCREEN-004 outputs + poster PNG (COMP-030, COMP-031) | Must | D6_Implementation_Plan |
| TASK-025 | SCREEN-005 manual form fallback (COMP-032) | Must | D6_Implementation_Plan |
| TASK-026 | SCREEN-007 found-child log + API-007 (COMP-034) | Must | D6_Implementation_Plan |
| TASK-027 | SCREEN-009 queue + API-008 (COMP-035, COMP-036) | Must | D6_Implementation_Plan |
| TASK-028 | SCREEN-010 match review + API-009/010 (COMP-037, COMP-038) | Must | D6_Implementation_Plan |
| TASK-029 | SCREEN-012 case detail, close, audit + API-011/013 (COMP-039, COMP-042) | Must | D6_Implementation_Plan |
| TASK-030 | SCREEN-011 map + table + API-012 (COMP-040, COMP-041) | Must | D6_Implementation_Plan |
| TASK-031 | SCREEN-008 sign-in + SCREEN-016 (COMP-047, COMP-045) | Must | D6_Implementation_Plan |
| TASK-032 | SCREEN-013 admin: redact identifying payloads + verify [MOCK] + API-016/017 (COMP-043) | Must | D6_Implementation_Plan |
| TASK-033 | Demo-safe cache + fixtures for 3 cases + cached badge (COMP-017, COMP-048) | Must | D6_Implementation_Plan |
| TASK-034 | Accessibility pass: axe + keyboard + 320 CSS px reflow, 200 % text zoom and 400 % viewport zoom | Must | D6_Implementation_Plan |
| TASK-035 | Bangla copy review; all empty/error/loading states | Must | D6_Implementation_Plan |
| TASK-036 | Test on 2 Android phones (Bangla rendering, mic, 360 px) | Must | D6_Implementation_Plan |

| ID | Label | Priority | Defined in |
|---|---|---|---|
| TASK-037 | Unit tests: normaliser, matcher, state guard, wording guard | Must | D6_Implementation_Plan |
| TASK-038 | Playwright E2E: essential journey + safeguarding TCs | Must | D6_Implementation_Plan |
| TASK-039 | Evaluation harness: smoke/dev run first; held-out run after manifest/model freeze | Must | D6_Implementation_Plan |
| TASK-040 | WER recordings (60 clips) + run | Must | D6_Implementation_Plan |
| TASK-041 | Timing study, ≥ 5 testers, assistant vs manual form | Must | D6_Implementation_Plan |
| TASK-042 | Production deploy + managed DB seed + smoke test | Must | D6_Implementation_Plan |
| TASK-043 | Final evaluation run on freeze build | Must | D6_Implementation_Plan |
| TASK-044 | Round 2 deck, 5–8 slides | Must | D6_Implementation_Plan |
| TASK-045 | Round 2 demo video, 1–2 min | Must | D6_Implementation_Plan |
| TASK-046 | README, Drive folder (links TXT, deck, video), reply to selection email | Must | D6_Implementation_Plan |
| TASK-047 | 13219 summary (FR-010) | Should | D6_Implementation_Plan |
| TASK-048 | Synthetic fixture import + API-015 (FR-024) | Should | D6_Implementation_Plan |
| TASK-049 | Edit after submit + API-005 + SCREEN-006 (FR-029) | Should | D6_Implementation_Plan |
| TASK-050 | Parent status API-018 (FR-030) | Should | D6_Implementation_Plan |
| TASK-051 | Duplicate review SCREEN-014 (FR-023) | Could | D6_Implementation_Plan |
| TASK-052 | Disaster mode SCREEN-015 (FR-036) | Could | D6_Implementation_Plan |
| TASK-053 | Finale local stack: Docker DB, model cache, fixtures, `DEMO_MODE=1` | Must | D6_Implementation_Plan |
| TASK-054 | 5 rehearsals, ≥ 1 internet-off | Must | D6_Implementation_Plan |
| TASK-055 | Finale deck, 5–10 min | Must | D6_Implementation_Plan |

| ID | Label | Priority | Defined in |
|---|---|---|---|
| TASK-056 | Fixes from mentor session 2 (12–13 Oct) | Should | D6_Implementation_Plan |
| TASK-057 | 5–10 user conversations, only if OQ-019 allows | Should | D6_Implementation_Plan |
| TASK-058 | Backup recording of the finale demo | Must | D6_Implementation_Plan |
| TASK-059 | `npm run spend` + spend cap check (DECISION-017) | Must | D6_Implementation_Plan |
| TASK-060 | Performance check against D2 Part B §12.1 | Must | D6_Implementation_Plan |
| TASK-061 | Apply V2 required fields, DTOs, token/ idempotency and referral/count semantics | Must | D6_Implementation_Plan |
| TASK-062 | V2 schema invariants, restricted roles and full related-payload redaction | Must | D6_Implementation_Plan |
| TASK-063 | Atomic paid-call counters/reservations, replay handling and price sheet | Must | D6_Implementation_Plan |
| TASK-064 | Identity-grouped manifest, no-match/ duplicate hard negatives, frozen metrics | Must | D6_Implementation_Plan |

# 8 Relationships

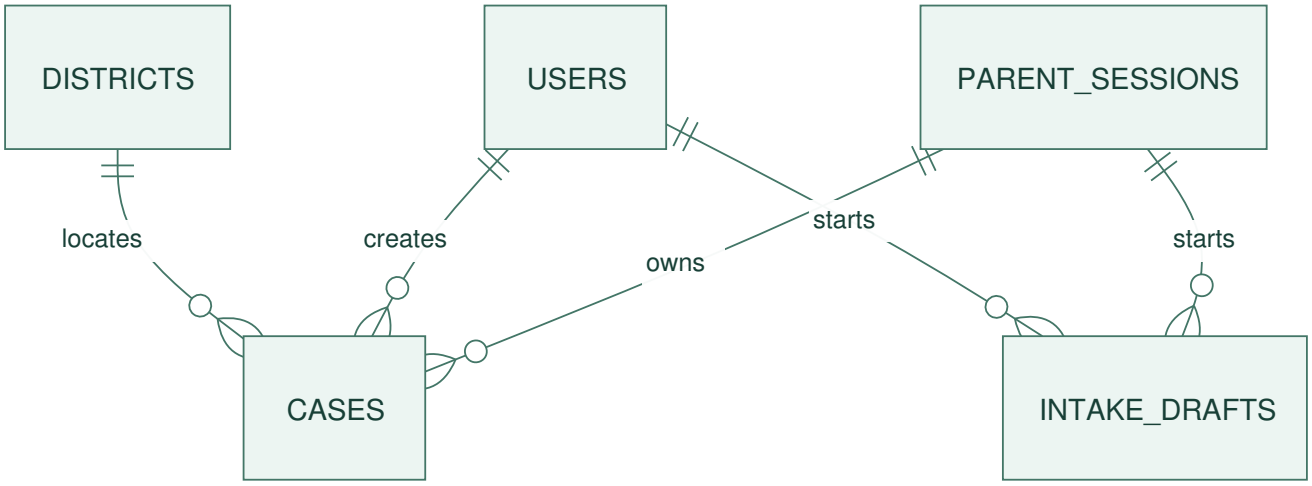
## 8.1 Link types

| From | To | Meaning | Where recorded |
|---|---|---|---|
| GOAL | STORY | Story contributes to goal | D1 §19 |
| STORY | FR | Requirement implements story | D1 §11 "Story" column |
| FR | SCREEN | Requirement visible on screen | D1 §10 stories, D3 §2 |
| FR / BR / NFR | TC | Test verifies requirement | D1 §11, §13.1; D6 §8 |
| BR | FR | Rule enforced by requirement | D1 §13.1 |
| FR | API | Endpoint serves requirement | D2 §5 |
| API | COMP | Component handles endpoint | D2 §4 |
| COMP | ENTITY | Component reads/writes table | D2 §4, §6 |
| SCREEN | COMP (UI) | Screen uses UI component | D3 §2, D4 §8 |
| SCREEN | ACT / D / E | Flow actions, decisions, errors | D3 §7, §9 |

| From | To | Meaning | Where recorded |
|---|---|---|---|
| TASK | FR / COMP / TC | Task builds and proves | D6 §5 |
| RISK | TASK / OQ | Mitigation and open question | D1 §16, D6, D7 |
| A | OQ | Assumption awaiting answer | D1 §14, §17 |

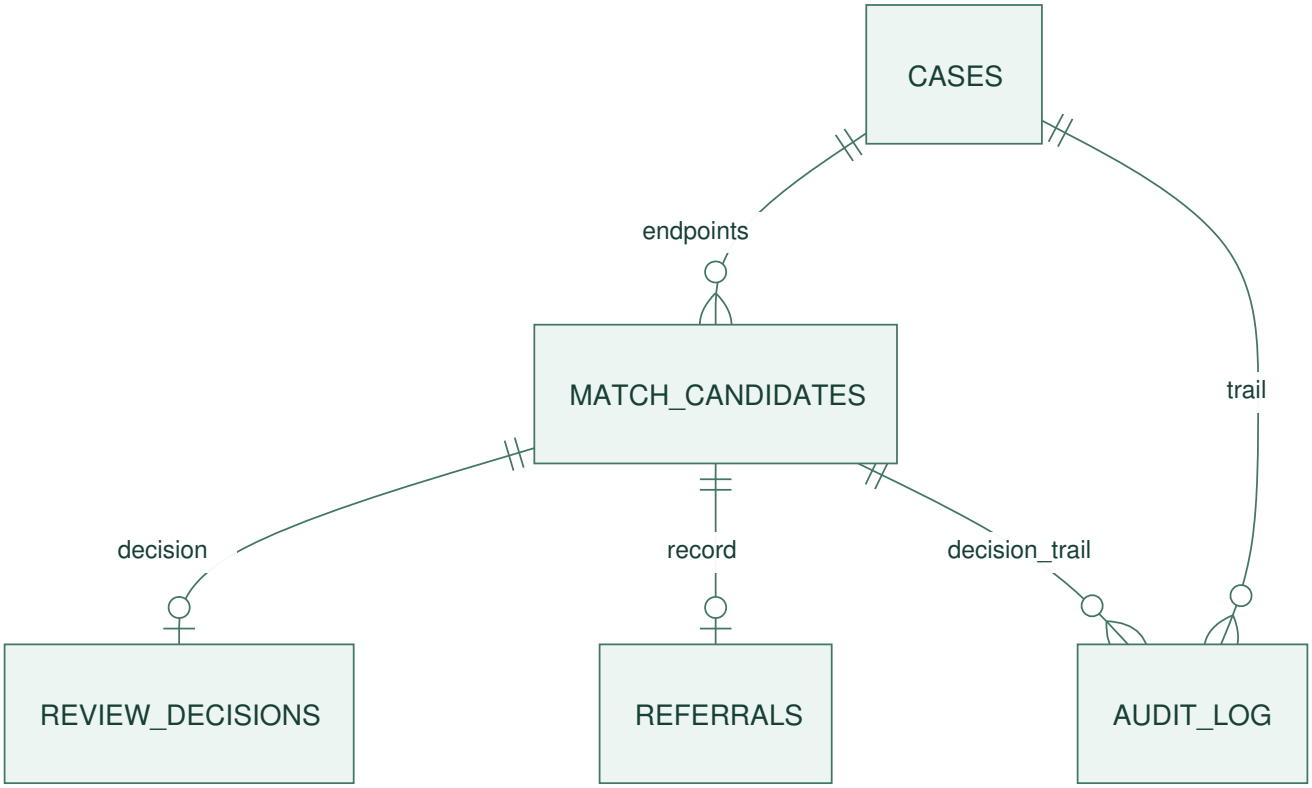
## 8.2 Domain ER diagram

### Intake and ownership relationships




V2 specification diagram · see adjacent rules for authority and evidence limits

## Candidate decision and audit relationships




V2 specification diagram · see adjacent rules for authority and evidence limits

| Additional relationship | Cardinality / rule |
|---|---|
| District → district_neighbours | One-to-many outgoing rows; no self-neighbour |
| Case → case_versions / generated_outputs | One-to-many; versioned snapshots/templates, cleared on redaction |
| Case ↔ duplicate_links | Each link references two ordered same-kind endpoints; reports retained |
| User → review_decisions / referrals | One-to-many authors; decision author must be active verified officer at decision time |
| User / parent_session → audit_log | Optional actor references; safe audit remains after payload redaction |
| Intake draft → ai_reservations / ai_calls | One-to-many metadata and usage links; no child payload in draft metadata |
| Reservation → ai_calls | At most one usage row per reservation |
| Case → ai_calls | Optional case link after submission |
| ai_budget singleton | Global reserved/committed cap enforced transactionally; no per-child identity |

## 8.3 Requirement traceability (essential journey)

| Goal | Story | Requirement / business rule | Screen | Acceptance |
|---|---|---|---|---|
| GOAL-001 reviewed report | STORY-001 voice | **FR-002** | **SCREEN-002** | **TC-001** |
| GOAL-001 reviewed report | STORY-004 review | FR-006; BR-004 review before register sharing | **SCREEN-003** | **TC-004** |
| GOAL-003 shortlist quality | STORY-009 candidates | FR-012 qualification/abstention | **SCREEN-009** | TC-009, TC-041 |
| GOAL-008 human decision | STORY-010 confirm | FR-016; BR-001 human review; BR-005 referral recorded | **SCREEN-010** | TC-010, TC-046/047 |
| GOAL-006 report status | STORY-013 district picture | FR-019; BR-014 counting basis | **SCREEN-011** | TC-012, TC-055 |

# 9 Glossary

| Bangla | English | Note |
|---|---|---|
| ফেরা | Fera ("return") | Product name |
| নিখোঁজ | Missing | Case kind |
| পাওয়া গেছে | Found | Case kind |
| সাধারণ ডায়েরি (জিডি) | General Diary (GD) | Police record |
| থানা | Thana (police station area) | Optional field |
| জেলা | District | 64 districts |
| শেষ দেখা | Last seen | Event time/place |
| খসড়া | Draft | Label on every output |
| পোস্টার | Poster | Text-only output |
| মিল | Match | Candidate pair |
| সম্ভাব্য মিল | Likely match | Queue label |
| মিলের মাত্রা | Similarity score | 0–100, not a probability |
| মিল নিশ্চিত করুন | Confirm match | Officer action |
| মিল নয় | Not a match | Officer action |
| সম্ভাব্য একই শিশু | Possible same child | Duplicate banner |
| রেফারেল | Referral | After confirm |

| Bangla | English | Note |
|---|---|---|
| শিশু সুরক্ষা সেবা | Child-protection services | Default referral agency text (OQ-018) |
| কর্মকর্তা | Officer | Role |
| স্বেচ্ছাসেবক | Volunteer | Staff persona |
| আশ্রয়কেন্দ্র | Shelter | Disaster mode |
| অজানা | Unknown | Empty field |
| সিনথেটিক | Synthetic | Seed data label |
| ক্যাশ | Cached | Demo-safe output label |
| পরিচয় তথ্য অপসারণ | Redact identifying payloads | Admin action |
| খোলা / যাচাই চলছে / বন্ধ | Open / Being checked / Closed | Parent status labels |
| পরিকল্পিত সম্প্রসারণ | Planned extension | Disaster mode banner |
| দুর্যোগ মোড | Disaster mode | Could |

# 10 V2 integrity and evidence

The companion code is generated once and embedded exactly. Registry counts and uniqueness are mechanically checked; definitions and references must be checked against the delivered source. Requirement-to-test links describe planned acceptance, not achieved application results. The verification folder distinguishes executed companion checks from pending application/PG16/host/partner evidence.

| Prefix | Count |
|---|---|
| A | 40 |
| ACT | 33 |
| API | 23 |
| BR | 26 |
| COMP | 50 |
| D | 16 |
| DECISION | 17 |
| DEP | 11 |
| E | 24 |
| ENTITY | 17 |
| FR | 37 |

| Prefix | Count |
|---|---|
| GOAL | 9 |
| NFR | 17 |
| OQ | 26 |
| PERSONA | 5 |
| RISK | 17 |
| SCREEN | 16 |
| STORY | 23 |
| TASK | 64 |
| TC | 66 |
| Total | 537 |

Authoritative locations: GOAL/PERSONA/OQ baseline in 00/D1; business requirements in D1; API/ENTITY in D2; SCREEN/ACT in D3; UI COMP in D4; shared interfaces and index here; TC/TASK execution specifications in D6. TASK-036 attribution is corrected. Reserved IDs are reservations, not duplicate definitions. The V2 delta adds API-021–023, ENTITY-015–017, TC-038–066 and TASK-061–064. Existing IDs remain stable.

## IDs introduced

- None. D5 is a registry only.

## IDs referenced

- All IDs listed in §7 (GOAL-001 – GOAL-009, PERSONA-001 – PERSONA-005, STORY-001 – STORY-023, FR-001 – FR-037, NFR-001 – NFR-017, BR-001 – BR-026, A-001 – A-040, DEP-001 – DEP-011, RISK-001 – RISK-017, OQ-001 – OQ-026, SCREEN-001 – SCREEN-016, COMP-001 – COMP-050, API-001 – API-020, ENTITY-001 – ENTITY-014, DECISION-001 – DECISION-017, E-001 – E-024, TC-001 – TC-037, ACT-001 – ACT-033, D-001 – D-016, TASK-001 – TASK-060)

↗ Return to contents